

MOHAMMED MUZAMMIL

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Education

Vidya Jyothi Institute of Technology, India - Bachelor of Technology (B.Tech) in Computer Science and Engineering (AI&ML) - CGPA: 8.73/10(Till 5th Sem) - Currently pursuing Seventh Semester	2022 - Present
Narayana Junior College, India (Class XII), Aggregate: 92.0%	2020 - 2022
Ravindra Bharathi School, India (Class X), CGPA: 10/10	2019 - 2020

Skills

Programming Languages: Python, Java, C,
Machine Learning & AI: Supervised Learning, Unsupervised Learning, Deep Learning, NLP
Tools & Frameworks: TensorFlow, PyTorch, Scikit-learn, OpenCV, MediaPipe, Git, Jupyter Notebook, VS Code
Databases:MySQL, PostgreSQL

Key Certifications

Startup India Learning Program (**Govt. of India**), Python Programming (**University of Michigan**), Supervised Machine Learning (**Stanford**), Unsupervised Learning & Reinforcement Learning (**Stanford**), Neural Networks (**DeepLearning.AI**), Cybersecurity Foundations (**Google**), Java & DBMS (**Oracle**).

Academic Projects

- AI-Driven Retrieval-Augmented Generation (RAG) System:**
Developed a state-of-the-art Retrieval-Augmented Generation (RAG) system that enables users to interact with and extract accurate, context-aware information from PDFs in real-time. The system integrates LangChain to build a robust RAG pipeline, while a seamless, user-friendly interface built with Streamlit allows for efficient querying and retrieval of answers to complex questions from local documents.
Technologies: LangChain, Streamlit, Python, AI, Natural Language Processing, Document Management Systems
- Gesture Mate:**
Developed an AI-driven gesture control system that enables precision mouse control through hand gestures. Utilized OpenCV and MediaPipe to create a real-time, zero-latency gesture recognition system with adaptive learning for personalized user experiences.
Technologies: OpenCV, MediaPipe, Python, Deep Learning
- Fuzz AI:**
Created a hybrid system that integrates machine learning with advanced fuzzing techniques to detect vulnerabilities in Open Source Software. The system continuously refines fuzzing strategies, learns from previous crashes, and retrains models to improve security, with detailed automated reports generated for insights.
Technologies: Machine Learning, Fuzzing, Python, TensorFlow, Scikit-learn
- HelioCharge AI** (Ongoing, Solo)
Developing an AI-powered solar charging station for electric vehicles, combining solar energy with intelligent energy management to optimize charging efficiency.
Technologies: AI, Solar Energy, Python, Energy Management Systems
- GIS & Remote Sensing** (Ongoing, in collaboration with **IIRS - Government of India**)
Working on AI-based projects for land use classification, traffic optimization, and 3D city modeling, leveraging machine learning and satellite imagery for geospatial analysis.
Technologies: AI, Machine Learning, Remote Sensing, GIS

International Conference on AI in Healthcare

- Member of a prestigious global conference**, engaging with world-renowned experts on AI-driven healthcare advancements.
- Interacted with Dr. Nageshwar Reddy (Padma Vibhushan awardee)** to explore AI applications in gastroenterology for early disease detection.
- Engaged with Rajiv Kumar (Managing Director, Microsoft India)** on AI's role in predictive healthcare analytics and decision support systems.
- Collaborated with Sravanthi Parsa (Vice Chair, ASGE AI in Gastroenterology)** to address challenges in AI adoption, contributing to discussions on "**Valley of Death in Medical Innovation: Overcoming Barriers to Reimbursement.**"

Research & Innovation in AI and Healthcare

AI-Enhanced Endoscopy for Real-Time Cancer Detection (Ongoing)

Developing an AI system to assist doctors by detecting cancerous lesions in real-time during endoscopic procedures. It includes:-

- Using **deep learning-based object detection** to identify abnormal tissues in live video feeds.
- Training the model on **medical imaging datasets (Kvasir)** to improve accuracy.
- Deploying on **low-power AI hardware** for real-time processing with minimal delay.
- Aiming to enhance **early cancer detection** and assist medical professionals in diagnosis.
- Exploring **quantum computing** to accelerate deep learning inference, enabling **near-instantaneous lesion detection** with unparalleled accuracy and computational efficiency, revolutionizing real-time medical diagnostics.

Visionary Leadership & High-Impact Recognitions

- **HACK Summit**

Actively contributed to the **HACK 2.0 Summit** by Hyderabad City Security Council where I had the esteemed privilege of interacting with **Sri. Vikram Singh Mann, IPS, Addl. Commissioner of Police (Law & Order)**. Our substantive discussion centered on proposing visionary frameworks for proactive cybersecurity defense, an exchange met with considerable approbation, underscoring its perceived strategic value in evolving contemporary public security paradigms.

- **Team Leader - National Hackathon**

Led a team to **victory** at the college-level hackathon, securing a spot for submitting the project at **National-Level Hackathon**. Played a key role in strategy, technical execution, and team coordination.

- **EDU Summit Representative**

Personally selected by the **Head of the Department** to represent and lead the department at the **EDU Summit (Fostering Innovation & Entrepreneurship through Industry Collaborations) at JNTU-H**, showcasing leadership and vision.

- **Literary Secretary**

Served as the Literary Secretary for **three consecutive years**, spearheading literary and cultural events, mentoring students in public speaking and creative writing, and fostering a culture of intellectual engagement and communication excellence.

- **School Pupil Leader (SPL)**

Led the entire student body as the **School Pupil Leader (SPL)**, overseeing school-wide initiatives, organizing events, and fostering leadership among peers. Played a pivotal role in student governance, ensuring smooth coordination between students and faculty.

Entrepreneurial Innovation Research(Ongoing)

AI-Driven Entrepreneurial Ecosystem Development for Emerging Markets: Working on developing an AI-driven entrepreneurial ecosystem tailored for emerging markets. This research focuses on utilizing AI to analyze and address the unique challenges faced by entrepreneurs in these regions. The goal is to create an adaptive and inclusive ecosystem that fosters innovation, provides tailored support, and enhances the success rate of startups.

Internship Experience

Goldman Sachs – Software Programming Virtual Program

Simulated IT security enhancements, identifying vulnerabilities using Hashcat and proposing advanced cybersecurity measures. Demonstrated strategic thinking in recommending modern security protocols.

Vision for Impact & Innovation

At the intersection of AI and entrepreneurship, I am committed to building technologies that don't just innovate but redefine industries. My mission is to create AI-driven solutions that transform patient outcomes, fortify cybersecurity, and drive sustainable advancements. With a relentless pursuit of excellence, I aim to bridge the gap between cutting-edge research and real-world impact because **True Innovation Is Not About Following Trends; It's about Setting Them.**